

SHiP Timing Bar:
From First Principles to Precision Timing
EJ-230 + Vikuiti ESR | Napkin First, Geant4 Second

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SHiP timing layer: reject scattered muon background from charm/tau decays. Target: $\sigma_t < 100$ ps.

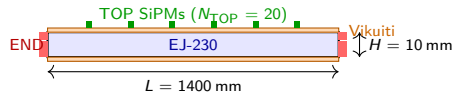
t_0 = instant of muon–bar interaction (“clock start”);

$\sigma_t = \sigma(\hat{t}_0 - t_0)$, reconstructed from SiPM hit times.

The detector bar: $1400 \times 60 \times 10$ mm³ EJ-230 scintillator.

Readout: END SiPMs (8 per face) + TOP SiPMs (N_{TOP} along the bar).

Reflector: Vikuiti ESR on all non-SiPM surfaces ($R = 0.95$).



$W = 60$ mm; 8 END SiPMs per face (6×6 mm²).

What is the “napkin”?

A first-principles back-of-envelope estimate using the bar geometry and the optical physics of TIR in a slab (Knoll, Ch. 8). It predicts trends analytically. Geant4 then confirms or reveals corrections.

For EJ-230 ($n = 1.58$, air $n_0 = 1$):

$$\theta_c = \arcsin\left(\frac{1}{1.58}\right) = 39.27^\circ$$

TIR fraction (slab geometry, Knoll Ch. 8 §III):

$$f_{\text{TIR}} \approx 2 \cos \theta_c - 1 = 0.548$$

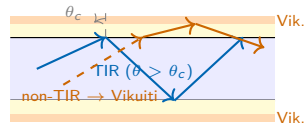
Photons hitting the $H = 10$ mm face at $\theta > \theta_c$ from the normal are totally reflected and guided toward the ends.

Per end (half travel rightward):

$$f_{\text{TIR/end}} = \frac{f_{\text{TIR}}}{2} = 0.274$$

54.8% of photons are TIR-trapped.

The remaining 45.2% escape through the H -faces but are recovered by the Vikuiti reflector (Population II).



Blue: TIR-guided (54.8%); *orange dashed:* non-TIR, recovered by reflector.

Population I: TIR-guided

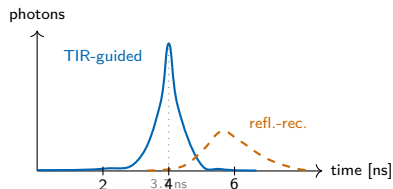
- Travel at zig-zag angle; never hit Vikuiti
- Path to END: 700 mm to 1106 mm
- Arrive 3.689 ns to 5.829 ns after emission
- These dominate END timing

Population II: Reflector-recovered

- Exit slab at $\theta < \theta_c$, hit Vikuiti
- Reflected ($R = 0.95$ per bounce) back into bar
- Re-enter at various angles; can become TIR
- Arrive *later* with more spread
- Account for 23% of extra Npe vs napkin

END estimator uses only the earliest few photons (m-average).

TOP estimator uses photons arriving at the TOP surface nearby — dominated by Population I with short path ~ 10 mm.



Schematic. END timing is set by the TIR-guided peak.

Bounces and Reflector Attenuation

TIR photons zig-zag at angle

$\alpha_c = 90^\circ - \theta_c = 50.73^\circ$ from the bar surface.

Number of bounces to traverse $L/2 = 700$ mm on $H = 10$ mm faces:

$$N_{\text{bounce}}^H = \frac{(L/2) \cdot \tan \alpha_c}{H} = \frac{700 \times 1.223}{10} \approx 85.6$$

For $W = 60$ mm faces: $N_{\text{bounce}}^W \approx 14.3$.

Reflector attenuation length:

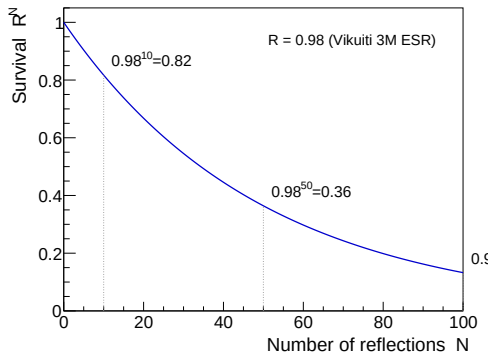
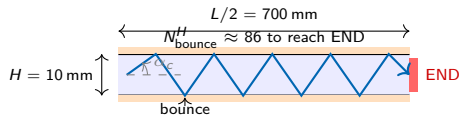
$$\Lambda_{\text{refl}} = \frac{H}{\tan \alpha_c \cdot \ln(1/R)} \approx 405 \text{ mm}$$

Napkin ($R = 0.98$): $\Lambda_{\text{refl}}^H \approx 405$ mm. Sim. data ($R = 0.95$):

$\Lambda_{\text{refl}}^H \approx 159.4$ mm.

At $L/2 = 700$ mm: survival = $0.95^{85.6} \approx 1.2\%$.

Most photons reaching the **END** did *not* bounce off the H -face Vikuiti.

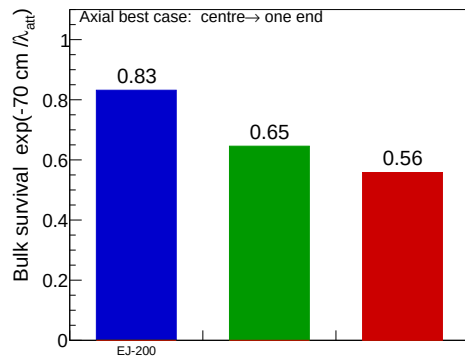


MIP through $H = 10$ mm EJ-230: $\Delta E = 2.0$ MeV.

Step	Photons	Factor
Produced	19 400	$Y \times \Delta E$
TIR toward 1 end	5 320	$f_{\text{TIR/end}}$
Bulk survival	2 969	$e^{-700/1200}$
At SiPM	1 425	cov. 0.48
Detected (PDE 0.40)	570	$\times 0.40$
Geant4	701	+23%

23% surplus = reflector-recovered photons.

Napkin: TIR-only path. G4 includes Population II.



EJ-200 EJ-204 EJ-230
Bulk survival $e^{-L/(2\lambda)}$ at $x = 0$ ($L/2 = 700$ mm).

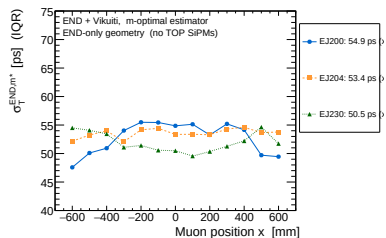
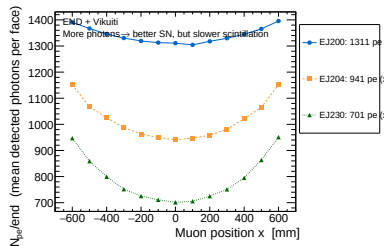
Material Choice: EJ-230 is the Fastest Scintillator Here

	EJ-200	EJ-204	EJ-230
λ_{att} [mm]	3800	1600	1200
τ_{decay} [ns]	2.1	1.8	1.5
τ_{rise} [ns]	0.9	0.7	0.5
Yield [ph/MeV]	10000	10400	9700
n	1.58	1.58	1.58
N_{pe}/end (G4)	1311	941	701
σ_t ($x=0$) [ps]	54.86	53.36	50.49[†]
σ_t (bar mean) [ps]	52.72	53.57	51.95

[†] GEN-3 END-only (scan_end_vikuiti), m^* -opt.

$x = 0$: EJ-230 (50.49 ps) < EJ-204 (53.36 ps) < EJ-200 (54.86 ps).

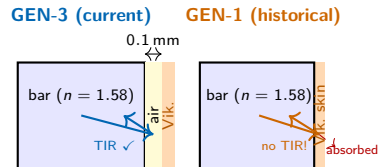
Bar mean: EJ-230 (51.95 ps) < EJ-200 (52.72 ps) < EJ-204 (53.57 ps).



Current geometry (GEN-3):

- Explicit **air gap** (0.10 mm) around bar
- Explicit **Mylar reflector** (0.05 mm) outside air
- Bar→air: `CreateBarSurface()` — `dielectric_dielectric`, polished → natural Fresnel + TIR for $\theta > \theta_c$
- Air→Mylar: `CreateBarSkinReflector()` as border surface — `dielectric_dielectric`, $R = 0.95$ constant
- SiPM: `dielectric_metal`, wavelength-dependent PDE (AFBR-S4N66P024M)

Key: the bar–air surface implements *natural* Fresnel TIR. No artificial reflectivity is set on the bar itself. Vikuiti only acts on photons that escape the bar.



GEN-1 skin surface on bar eliminates TIR entirely → photon deficit ($\times 1500$, see B2).

Photon yield at $x = 0$, EJ-230:

Path	Napkin	G4 (END-only)
TIR-guided / end	570	—
Reflector-recovered	0	—
Total / end	570	701.3

$\Delta N_{pe}/N_{nap} = +23\%$ (END-only).

Napkin accuracy: -17% (predicted deficit).

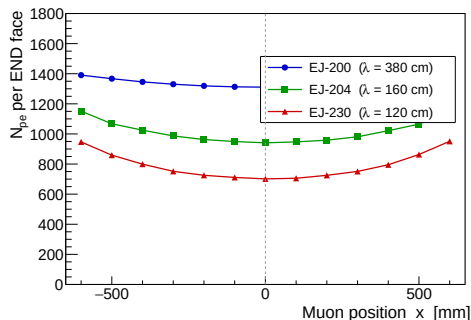
Reason: napkin counts only the TIR-guided population. Reflector-recovered photons add a systematic $+23\%$ that the napkin cannot predict without knowing R .

Timing estimate (napkin, order-of-magnitude):

$$\sigma_t \approx \frac{\tau_{\text{decay}}}{\sqrt{N_{pe}}} = \frac{1500 \text{ ps}}{\sqrt{570}} \approx 63 \text{ ps}$$

Geant4 (hybrid geometry, $N_{\text{TOP}} = 20$, $m = 8$):

$\sigma_{\text{END}} = 53.68 \text{ ps}$ — within 20%. END-only ($N_{\text{TOP}} = 0$, $m^* = 7$): 50.49 ps (scan_end_vikuiti).



N_{pe}/end vs x for three scintillators. EJ-200 highest, EJ-230 lowest but fastest decay.

Definition:

$$T_0^{\text{END}} = \frac{1}{2} \left(\bar{t}_L^{(m)} + \bar{t}_R^{(m)} \right), \quad \bar{t}^{(m)} = \frac{1}{m} \sum_{i=1}^m t_i^{(\text{sorted})}$$

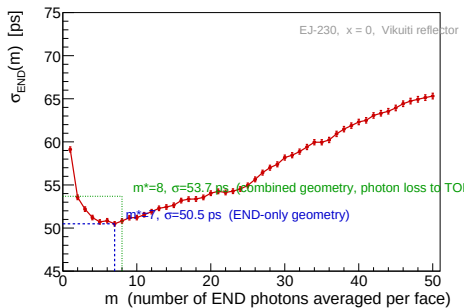
Average of first m photon arrival times per END face.

Optimal m : balance between noise rejection (large m) and avoiding slow photons (small m). Oracle: $m^* = 8$ at $x = 0$.

Result at $x = 0$, $m = 8$:

$$\sigma_{\text{END}} = 53.68 \text{ ps}$$

Path to END: $L/2 = 700 \text{ mm}$ axial (3.689 ns) up to $(L/2)/\sin \theta_c = 1106 \text{ mm}$ at θ_c from the H -face normal (5.829 ns).



σ_{END} vs m at $x = 0$. Minimum near $m = 8$.

Definition:

$$T_0^{\text{TOP}} = t_{(k)}^{\text{pool}}$$

k -th smallest arrival time among *all* TOP SiPMs pooled together.

Key difference from END: TOP SiPMs sit along the bar; a hit on the nearest SiPM travels only ~ 10 mm.

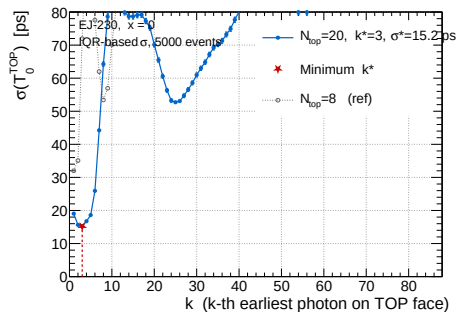
$$\tau_{\text{prop}}^{\text{TOP}} \approx \frac{10 \text{ mm}}{v_g} = 53 \text{ ps}$$

vs. $\tau_{\text{prop}}^{\text{END}} \in [3.689 \text{ ns}, 5.829 \text{ ns}]$.

Result at $x = 0$, $k = 3$, $N_{\text{TOP}} = 20$:

$$\sigma_{\text{TOP}} = 15.20 \text{ ps}$$

3.5 $\times$ better than END. The key is proximity, not photon statistics.



σ_{TOP} vs k at $x = 0$, $N_{\text{TOP}} = 20$. Minimum at $k = 3$.

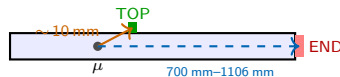
Propagation jitter — not photon counting — limits END timing.

	END	TOP
Path length	700 mm–1106 mm	~ 10 mm
τ_{prop}	3.689 ns–5.829 ns	~ 53 ps
Path dispersion	large	negligible
N_{pe} used	~ 8	~ 3 (pooled)
σ_t at $x = 0$	53.7 ps	15.2 ps

$m = 8$ average uses earliest photons, near the axial bound (3.689 ns).

The TOP SiPM nearest to the muon hit position receives photons that have traveled almost no distance. The 3rd-order statistic (out of ~ 1300 pooled photons) suppresses random single-photon fluctuations.

The factor $3.5\times$ improvement is purely from geometry, not hardware.



The nearest TOP SiPM is at most 35 mm from any muon hit ($N = 20$).

From END $\Delta t = t_R^{(m)} - t_L^{(m)}$ vs hit position x :

$$\langle \Delta t(x) \rangle = a + \frac{x}{v_{\text{eff}}/2}$$

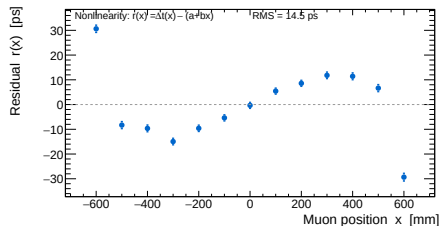
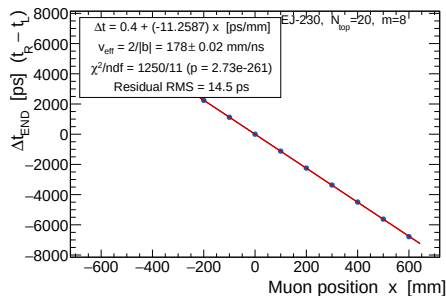
Fit ($m = 8$, EJ-230, $N_{\text{TOP}} = 20$):

- $v_{\text{eff}} = 177.6 \text{ mm/ns}$
- $a = 0.44 \pm 0.38 \text{ ps}$
- $\chi^2/\text{ndf} = 1250/11$, $p \approx 0$
- Residual RMS = 14.5 ps

Highly significant nonlinearity. S-shape pattern driven by photon pool edge asymmetry.

Compare: $v_g = 189.7 \text{ mm/ns}$; $v_{\text{eff}} < v_g$ because $m = 8$ average includes photons that bounced more.

Position resolution: $\sigma_x = \frac{v_{\text{eff}}}{2} \sigma_{\Delta t} = 7.9 \text{ mm}$.



BLUE (Best Linear Unbiased Estimator):

$$T_0^{\text{BLUE}} = w_{\text{END}} T_0^{\text{END}} + w_{\text{TOP}} T_0^{\text{TOP}}$$

Weights from covariance matrix ($\sigma^2 = \sigma_{\text{IQR}}^2$, $C_{ET} = \text{Cov}(T_E, T_T)$ empirical):

$$w_{\text{END}} = \frac{\sigma_T^2 - C_{ET}}{\sigma_E^2 + \sigma_T^2 - 2 C_{ET}}$$

At $x = 0$, $k = 3$, $m = 8$, $N_{\text{TOP}} = 20$:

Quantity	Value
σ_{END}	53.68 ps
σ_{TOP}	15.20 ps
C_{ET}	196.3 ps ²
w_{END}	$\lesssim 0.1$ (not resolved, $n = 5000$)
σ_{BLUE}	15.21 ps

$\sigma_{\text{BLUE}}^{\text{formula}} (w_{\text{END}} = 0.013, \text{mixed cov.}) = 15.18 \text{ ps}$; $\sigma_{\text{IQR}}^{\text{event}} = 15.21 \text{ ps}$.

Different scale metrics of a heavy-tailed distribution — not a consistency check.

Why does TOP dominate?

$\sigma_{\text{TOP}}^2 \ll \sigma_{\text{END}}^2$: the END estimator is so much worse that it contributes almost nothing to the BLUE combination.

The BLUE estimator is, to excellent approximation, just the TOP estimator.

σ_{BLUE} and σ_{TOP} differ by less than one bootstrap standard error ($\approx 0.17 \text{ ps}$ at $n = 5000$).

w_{END} **not resolved** at $n = 5000$: IQR flat to within errors over $w \in [0, 0.1]$ (total variation 0.26 ps). See App. G2.

Fixed parameters across all positions — deployable in real detector.

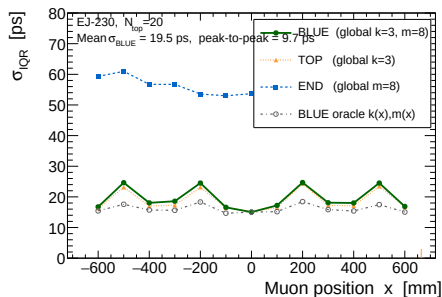
x [mm]	σ_E	σ_T	σ_B	w_E
-600	59.3	16.3	16.7	0.115
-500	61.0	23.2	24.6	0.208
-400	56.7	17.0	18.1	0.178
-300	56.7	17.3	18.6	0.183
-200	53.6	23.2	24.5	0.237
0	53.7	15.2	15.0	0.086
+600	61.0	16.8	16.9	0.118

All σ in ps.

$N_{\text{TOP}} = 20$, EJ-230. $\dagger w_E$ computed position-by-position with `np.var()`; best-estimate IQR-based $w_E \lesssim 0.1$ at $x = 0$ (not resolved at $n = 5000$, see App. G2).

Peaks at $x = \pm 200, \pm 500$ mm: $k = 3$ is locally suboptimal there. Mean $\sigma_{\text{BLUE}} = 19.5$ ps across bar.

$\sigma_B(x=0) = 15.0$ ps (`v5`, `np.var` weights) vs 15.21 ps (best-estimate IQR weights); definitional, see App. G2.



Blue = global ($k = 3$, $m = 8$); gray = oracle ($k^*(x)$, $m^*(x)$).

Oracle: locally optimal $(k^*(x), m^*(x))$ — requires knowing muon position. Not deployable, but a diagnostic lower bound.

	Global	Oracle
σ_{BLUE} at $x = 0$	15.0	15.0
Mean σ_{BLUE}	19.5	16.1
Min σ_{BLUE}	15.0	14.7
Max σ_{BLUE}	24.7	18.5
Gap (mean)	3.4 ps	

$N_{\text{TOP}} = 20$, EJ-230.

The 3.4 ps gap is the cost of deploying fixed $k = 3$, $m = 8$ parameters across all positions.

At $x = 0$: global = oracle ($k^* = 3$, $m^* = 8$).

Peaks at $x = \pm 200$ mm ($k^* = 2$) and $x = \pm 500$ mm ($k^* = 1$): global $k = 3$ is suboptimal there.

$\sigma_B(x=0) = 15.0$ ps (v5, np.var weights) vs 15.21 ps (best-estimate IQR weights); definitional, see App. G2.

Oracle $k^*(x)$:

x [mm]	(k^*, m^*)
−600	(2, 10)
−500	(1, 11)
0	(3, 8)
+500	(1, 5)
+600	(2, 5)

$k^* = 1$ (first photon) at bar edges where TOP photons arrive sparsely. $k^* = 3$ at bar center where N_{TOP} pool is densest.

Each N_{TOP} value uses its own simulated file with SiPMs physically placed at correct positions. Not software ablation.

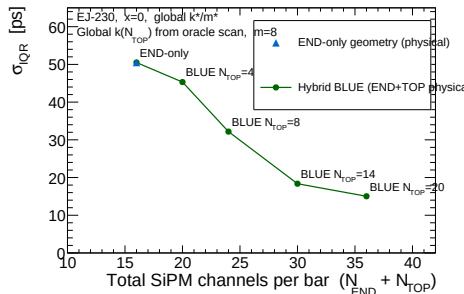
N_{TOP}	N_{ch}	k^*	σ_{TOP}	σ_{BLUE}
4	20	1	78.2	45.3
8	24	1	32.0	32.2
14	30	2	18.7	18.4
20	36	3	15.2	15.0

All σ in ps. $m = 8$,

EJ-230, $x = 0$. $N_{\text{ch}} = 16 + N_{\text{TOP}}$.

END-only ($N_{\text{TOP}} = 0$): $\sigma \approx 50.5$ ps.

Adding TOP SiPMs: $50.5 \rightarrow 45.3 \rightarrow 32.2 \rightarrow 18.4 \rightarrow 15.0$ ps.



σ_{BLUE} vs N_{ch} . Elbow at $N_{\text{TOP}} = 14$ ($N_{\text{ch}} = 30$). 20 TOP channels for last 3.4 ps gain (unrelated to the S17 oracle gap).

END Δt method:

$$x_{\text{rec}} = \frac{v_{\text{eff}}}{2} \cdot \Delta t, \quad \sigma_x = \frac{v_{\text{eff}}}{2} \sigma_{\Delta t}$$

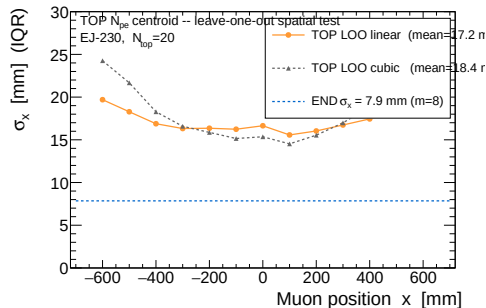
At $m = 8$: $\sigma_x = 7.9 \text{ mm}$ with $v_{\text{eff}} = 177.6 \text{ mm/ns}$.
No edge bias. Linear calibration.

TOP N_{pe} centroid (LOO):

$$x_{\text{centroid}} = \frac{\sum_i x_i N_i}{\sum_i N_i}$$

Leave-one-out cross-validation on 12/13 positions.
Result: $\sigma_x = 17.2 \text{ mm}$ (linear fit), **+29 mm bias at bar ends.**

END method is 2× better with no edge bias.



TOP centroid LOO: large edge bias at $x = \pm 600 \text{ mm}$.
 $x_{\text{true}} = \pm 600 \text{ mm} \rightarrow x_{\text{centroid}} \approx \pm 313 \text{ mm}$ (saturation).

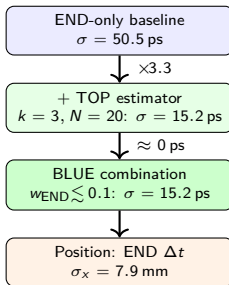
Selected configuration:

- Scintillator: **EJ-230** ($\tau = 1.5$ ns, $\lambda_{\text{att}} = 1200$ nm)
- Reflector: **Vikuiti ESR** ($R = 0.95$ in data; code 0.98 since c7acb7a)
- END SiPMs: **8 per face** = 16 total
- TOP SiPMs: **20 along bar** (step ≈ 70 mm)
- Total channels: **36**

Performance (global $k = 3$, $m = 8$):

Quantity	Value
σ_t at $x = 0$	15.2 ps
Mean σ_t across bar	19.5 ps
σ_x (END Δt)	7.9 mm
N_{pe}/end at $x = 0$	514.9 (hybrid, $N_{\text{TOP}} = 20$)

Target: $\sigma_t < 100$ ps. **Achieved: 15.2 ps intrinsic (see App. I2) — 6.6× better than target.**

Path to this result:

$\times 3.3$ = TOP geometry (near-field path).
BLUE: essentially free combination.

Napkin first — Geant4 second

First-principles predictions (TIR, bounce counting, photon budget) correctly set the order of magnitude and explain the trends. Geant4 confirms quantitatively and reveals the reflector-recovered contribution.

Optical physics:

- TIR traps 54.8% of photons toward ends
- Vikuiti ($R=0.95$) recovers non-TIR photons (+23%)
- EJ-230 is fastest despite lower yield

Estimators:

- END m-avg ($m = 8$): $\sigma = 53.7$ ps
- TOP k -th order ($k = 3$, $N = 20$):
 $\sigma = 15.2$ ps
- BLUE: $\sigma = 15.2$ ps, $w_{\text{END}} \lesssim 0.1$ (END contributes marginally)

Optimisation:

- Global ($k = 3$, $m = 8$): mean 19.5 ps, deployable
- Oracle lower bound: mean 16.1 ps (-3.4 ps gap)
- Pareto elbow at $N_{\text{TOP}} = 14$; $N = 20$ for best performance

Position:

- END Δt : $\sigma_x = 7.9$ mm, no edge bias
- TOP centroid: $\sigma_x = 17.2$ mm, +29 mm edge bias

Design: EJ-230, 36 channels (16 END + 20 TOP), $\sigma_t = 15.2$ ps.

Appendix

Technical backup slides

CreateBarSurface() — bar $\rightarrow$ air border:

- Type: `dielectric_dielectric`
- Model: `unified`; Finish: `polished`
- No artificial reflectivity — pure Fresnel
- TIR automatic for $\theta > \theta_c = 39.3^\circ$

CreateBarSkinReflector() — air $\rightarrow$ Mylar border:

- Type: `dielectric_dielectric`
- Model: `unified`; Finish: `polished`
- $R = 0.95$ constant (wavelength-independent)
- Used as *border* surface (naming is legacy from skin-surface era)

CreateSiPMSurface() — SiPM sensitive:

- Type: `dielectric_metal`
- PDE: wavelength-dependent (AFBR-S4N66P024M curve)
- Mean PDE over detected photons: 0.607 (selection-biased)

A2 — $R=0.95$ vs $R=0.98$

All simulation data shown: $R = 0.95$ (Materials.cc at data generation).

Code corrected to $R = 0.98$ in commit `c7acb7a` (2026-09-01); re-simulation pending.

Napkin uses $R = 0.98$ for reflector attenuation estimates ($\Lambda_{\text{refl}}^H \approx 405$ mm).
With $R = 0.95$ (simulation data):
 $\Lambda_{\text{refl}}^H = 159.4$ mm (shorter — more loss).

GEN-3 volume hierarchy:

- WorldLV $\supset$ BarLV (EJ-230)
- BarLV surrounded by AirGapLV (0.10 mm each side)
- AirGapLV surrounded by MylarLV (0.05 mm)
- Border surfaces:
 - scintAirSurface: BarLV $\rightarrow$ AirGapLV
 - airReflectorSurface: AirGapLV $\rightarrow$ MylarLV

This geometry correctly separates:

- 1 Fresnel refraction at bar–air interface (natural physics)
- 2 Vikuiti reflectivity at air–Mylar interface ($R = 0.95$)

GROUPVEL verification: propagation time in air is set correctly (no superluminal photons).

A4 — PDE Clarification

Mean PDE over *detected* photons: 0.607

This is **not** the detection fraction. It is the mean PDE weighted by which photons arrived at a SiPM — a selection-biased estimator.

Detection fraction = $N_{\text{det}}/N_{\text{emit}}$ (not directly available from current output).

Photon budget uses $\text{PDE}_{\text{napkin}} = 0.40$ (assumed flat, no selection bias).

Coverage fraction:

$$f_{\text{cov}} = \frac{8 \times 36 \text{ mm}^2}{60 \times 10 \text{ mm}^2} = 0.48$$

	GEN-1	GEN-3 END-only (scan_end_vikuiti)	GEN-3
Build	end-only (early)	build_end_vikuiti	build_end_vikuiti
Air gap	No	Yes (0.10 mm)	Yes (0.10 mm)
Reflector	dielectric_metal skin	border surface, $R=0.95$	border surface, $R=0.95$
TIR	Eliminated by skin	Natural Fresnel	Natural Fresnel
N_{pe}/end	~ 0.37 (BUG)	701.3	514.9
σ_{END}	N/A (insufficient pe)	50.49 ps ($m^* = 7$)	53.68 ps ($m = 8$)
TOP	No	No ($N = 0$)	Yes ($N = 4, 8, 14, 20$)
Status	INVALID	superseded	current

GEN-3 END-only (scan_end_vikuiti, $m^* = 7$, $x = 0$): $\sigma_{END} = 50.49$ ps; GEN-3 with $N_{TOP} = 20$: $\sigma_{END} = 53.68$ ps (~ 3 ps more). *Hypothesis*: TOP SiPMs intercept early photons that would otherwise reach the END face; not directly demonstrated (the comparison conflates N_{TOP} and estimator parameters simultaneously).

Symptom: GEN-1 simulations reported ~ 0.37 pe/event at END faces.

Cause: `dielectric_metal` skin surface placed directly on the bar volume. This surface type in Geant4 eliminates all internal reflections — photons cannot undergo TIR because the “metal” boundary absorbs at each encounter.

Effect:

- Without TIR, only photons emitted in the escape cone ($f_{\text{esc}} \approx 11\%$) can reach the END face
- Most of those are absorbed by the “Vikuiti” metal surface before reaching END
- Vikuiti/TIR ratio of $\sim 10^3$ claimed from these data was a simulation artifact

Fix (exec21-optfix): Changed to `dielectric_dielectric` border surface with $R = 0.95$, preserving natural TIR at bar–air interface.

Sanity check:

Napkin prediction: $N_{\text{pe}} = 570$ (GEN-3 agreement: 701).

GEN-1 claim: $N_{\text{pe}} \approx 0.37$ — factor $\sim 1500\times$ below napkin.

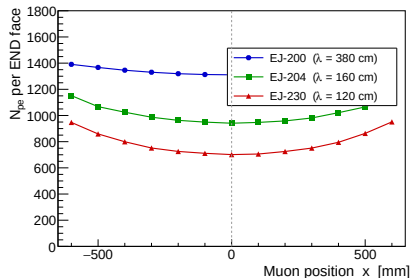
No physical mechanism can explain a factor-1500 deficit in a working scintillator.

The anomaly was a Geant4 surface type error, not a physics effect. All datasets from GEN-1 are invalid for physics analysis and must not be cited.

Property	EJ-200	EJ-204	EJ-230
λ_{att}	3800 mm	1600 mm	1200 mm
τ_{rise}	0.9 ns	0.7 ns	0.5 ns
τ_{decay}	2.1 ns	1.8 ns	1.5 ns
Yield	10000 ph/MeV	10400 ph/MeV	9700 ph/MeV
n	1.58	1.58	1.58
N_{pe}/end (G4, $x = 0$)	1311	941	701.3
N_{pe}/end (napkin)	—	—	570
σ_t (G4, $x=0$, m^* -opt)	54.86 ps	53.36 ps	50.49 ps
Napkin bulk survival	0.832	0.646	0.558

Note: σ_t values from GEN-3 END-only geometry (scan_end_vikuiti, m^* -optimal per position). GEN-3 with TOP SiPMs gives $\sigma_{\text{END}} = 53.68$ ps for EJ-230 at $x = 0$ (see App. B1).

Stage	Value	Note
Photons produced	19 400	$Y \times \Delta E$
TIR frac. (slab)	0.548	$2 \cos \theta_c - 1$
Toward one end	0.274	$\div 2$
TIR/end	5 320	$\times 0.274$
Bulk survival	0.558	$e^{-700/1200}$
After bulk	2 969	
SiPM cov.	0.480	288/600
At SiPM	1 425	
PDE	0.400	assumed
Napkin/end	570	TIR-only
G4/end (END-only)	701.3	+refl.-rec.
G4/end (hybrid, $N_{\text{TOP}}=20$)	514.9	TOP intercept
TOP interception	-26.6%	vs END-only
Surplus (END-only)	+23%	Pop. II



N_{pe} vs x , three materials. Dip at center.

x [mm]	σ_E [ps]	σ_T [ps]	σ_B [ps]	w_E	w_T	ρ
-600	59.3	16.3	16.7	0.115	0.885	0.088
-500	61.0	23.2	24.6	0.208	0.792	0.095
-400	56.7	17.0	18.1	0.178	0.822	0.098
-300	56.7	17.3	18.6	0.183	0.817	0.090
-200	53.6	23.2	24.5	0.237	0.763	0.086
-100	53.0	16.4	16.6	0.166	0.834	0.070
0	53.7	15.2	15.0	0.086	0.914	0.160
+100	53.8	16.5	17.2	0.166	0.834	0.072
+200	55.0	24.3	24.7	0.229	0.771	0.135
+300	55.3	17.2	18.1	0.185	0.815	0.087
+400	57.6	17.1	18.0	0.165	0.835	0.083
+500	59.7	23.5	24.5	0.214	0.786	0.089
+600	61.0	16.8	16.9	0.118	0.882	0.082

$k = 3$, $m = 8$, $N_{\text{TOP}} = 20$, EJ-230. $\sigma = \sigma_{\text{IQR}}$. w_E and ρ : v5 (np.var weights); $\rho = C/\sqrt{\text{var}_E \text{var}_T}$ (empirical corrccoef, *not* the BLUE-consistent $C/(\sigma_E \sigma_T) = 0.241$ at $x = 0$). See App. G1.

Two values of v_{eff} appear in the literature from this work:

- $v_{\text{eff}} = 184.6 \text{ mm/ns}$: from napkin/legacy analysis using near-first-photon estimator
- $v_{\text{eff}} = 177.6 \text{ mm/ns}$: from v5 analysis using $m = 8$ average estimator

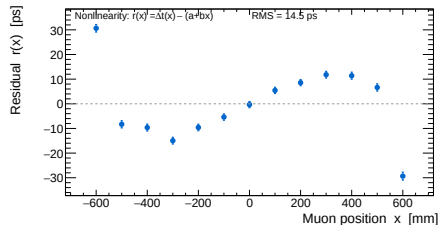
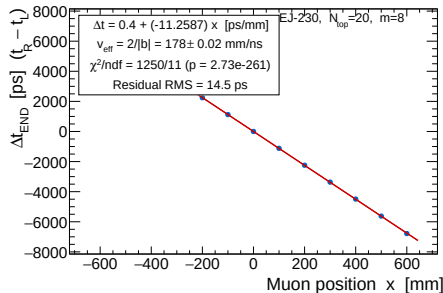
Both are correct for their respective estimators.

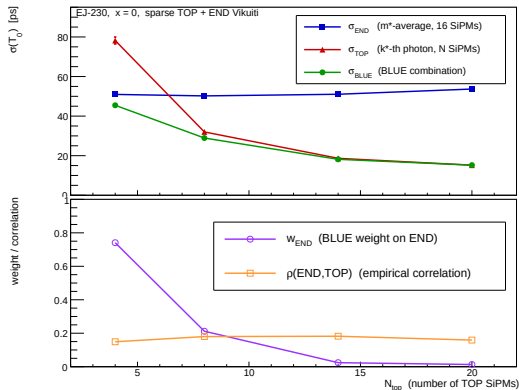
The $m = 8$ average includes photons that arrived slightly later (more bounces), so the effective propagation velocity is lower than for the first photon.

Compare: $v_g = 189.7 \text{ mm/ns}$.

$$v_g > v_{\text{eff}}^{(1)} > v_{\text{eff}}^{(m=8)}$$

Position resolution depends on which v_{eff} and which $\sigma_{\Delta t}$ are used consistently.





σ_{TOP} vs k for multiple positions. At bar center $k^* = 3$; at edges $k^* = 1$ or 2.

Why does k^* vary with x ?

At bar center ($x = 0$): the nearest TOP SiPM is always close (~ 35 mm). Larger k averages out fluctuations $\rightarrow k^* = 3$.

At bar edges ($x = \pm 600$ mm): the muon is near the end SiPMs but far from most TOP SiPMs. The TOP photon pool is sparser. The first photon ($k = 1$) carries the timing signal cleanly; waiting for the $k = 3$ photon adds slow outliers $\rightarrow k^* = 1$ or 2.

Global choice $k = 3$ performs well at center but suboptimally at edges (spikes at $x = \pm 200, \pm 500$ mm in $\sigma(x)$ profile).

BLUE formula:

$$w_E = \frac{\sigma_T^2 - C_{ET}}{\sigma_E^2 + \sigma_T^2 - 2C_{ET}}$$

$\sigma^2 = \sigma_{\text{IQR}}^2$ (diagonal); $C_{ET} = \text{Cov}(T_E, T_T)$ (best_est).

At $x = 0$: $\sigma_E = 53.68$ ps, $\sigma_T = 15.20$ ps,
 $C_{ET} = 196.3$ ps².

$C_{ET}/(\sigma_E\sigma_T) = 0.241$ (BLUE-consistent ρ ; *not* the
 corrccoef 0.16).

$$w_E = \frac{15.20^2 - 196.3}{53.68^2 + 15.20^2 - 2 \times 196.3} = 0.013$$

One defensible convention — see G2 for ambiguity.

Formula vs. event:

$\sigma^{\text{formula}} = 15.18$ ps (Gaussian BLUE algebra);
 $\sigma_{\text{IQR}}^{\text{event}} = 15.21$ ps (IQR of BLUE series).

Different scale metrics; agreement here is coincidental
 for non-Gaussian data.

Why $w_E \approx 0.013$?

$\sigma_E = 53.68$ ps $\gg$ $\sigma_T = 15.20$ ps:

END dominated by propagation jitter;
 almost all BLUE weight falls on the precise
 TOP estimator.

 ρ distinction:

$\rho_{\text{BLUE}} = 0.241 = C_{ET}/(\sigma_E^{\text{IQR}}\sigma_T^{\text{IQR}})$
 enters the BLUE denominator.

$\rho_{\text{emp}} = 0.160$ is the Pearson corrccoef
 of (T_E, T_T) residuals.

They differ because different σ estimates
 enter the two ratios.

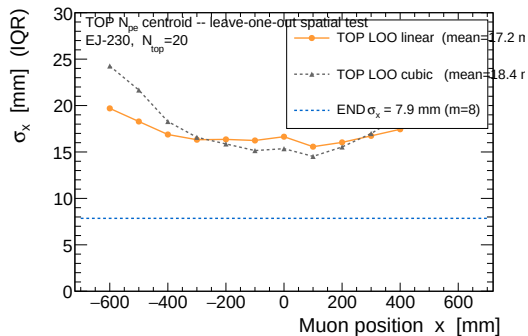
Three conventions for C_{ET} and w_E at $x = 0$:

Convention	C_{ET} (ps ²)	w_E	$\sigma_{\text{IQR}}^{\text{ev}}$ (ps)
best_est (IQR σ^2 , emp. cov)	196.3	0.013	15.21
robust IQR (polarization)	80.5	0.051	15.12
empirical (np.var + np.cov)	196.3	0.086	15.03

All three within 0.18 ps of each other — comparable to bootstrap SE (≈ 0.17 ps, $n = 5000$). The convention is not resolved by this dataset; no design conclusion depends on it.

$\sigma_{\text{IQR}}(w)$ flat to within one SE over $w \in [0, 0.1]$ (total variation 0.26 ps).

Qualitative BLUE check: w^* increases with σ_T as expected (w_E monotone increasing in σ_T^2 in BLUE). Empirical w^* (biased argmin, 81-point scan on the *same* $n=5000$ sample, $\lesssim 1.25\sigma$ improvement, **not resolved** — reported as a diagnostic, not a result): $x=+600$ ($\sigma_T=16.8$ ps, $w^*=0.065$), $x=0$ ($\sigma_T=15.20$ ps, $w^*=0.080$), $x=+200$ ($\sigma_T=24.3$ ps, $w^*=0.125$). Mostly monotone in σ_T ; $x=0$ vs $x=+600$ reversed — statistical noise.



LOO calibration: 12 training positions, 1 test. Linear and cubic fits shown.

Why does the centroid saturate?

The TOP SiPMs are uniformly distributed over ± 700 mm. When the muon hits at $x = +600$ mm, nearly all photons go to the rightmost SiPMs but the centroid cannot exceed the physical SiPM span.

True $x = \pm 600$ mm $\rightarrow$ centroid $\approx \pm 313$ mm.
Factor ~ 2 compression.

Leave-one-out results:

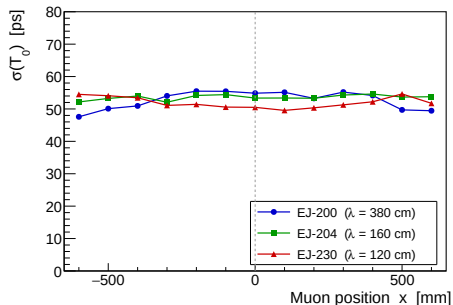
Fit	σ_x [mm]	Max $ b $ [mm]
Linear	17.2	28.8
Cubic	18.4	23.8

Cubic reduces edge bias but increases mean σ_x .

END Δt : $\sigma_x = 7.9$ mm, zero edge bias. Use END for position.

Sources of σ_t :

- **Photon emission timing:** convolution of $\tau_{\text{rise}} = 0.5 \text{ ns} + \tau_{\text{decay}} = 1.5 \text{ ns} \rightarrow$ time-to-first-photon distribution
- **Propagation jitter:** path length spread (TIR zig-zag + reflector-recovered)
- **SiPM SPTR:** datasheet (DS105) does not tabulate SPTR. Published measurement, optically identical $6 \times 6 \text{ mm}^2$ NUV-MT (AFBR-S4N66P014M; Lee et al., IEEE TRPMS 9(4), 406–411, 2025, DOI: 10.1109/TRPMS.2024.3518479):
intrinsic: $137 \pm 4 \text{ ps FWHM} \Rightarrow \sigma = \text{FWHM}/2.355 \approx 58 \text{ ps}$
w/ electronics: $172 \text{ ps FWHM} \Rightarrow \sigma \approx 73 \text{ ps}$
 $V_b = 48 \text{ V}$ ($\approx 15.5 \text{ V OV}$, near abs. max. of 16 V); SPTR degrades below $\approx 40 \text{ V}$. 48 V is the assumed operating point, not guaranteed.
 $\sigma_{\text{TOP}} = 15.2 \text{ ps}$ is a std. dev.; SPTR FWHM must be converted ($\sigma = \text{FWHM}/2.355$) before combining.
- **Electronic:** TDC binning, cable, trigger — not included in this simulation



σ_t vs position for three materials. Napkin: EJ-230 fastest at all positions (faster τ). G4 ($x = 0$, m^* -opt, GEN-3 END-only): EJ-230 (50.49 ps) < EJ-204 (53.36 ps) < EJ-200 (54.86 ps).

Current simulation includes: photon emission + propagation + SiPM PDE.

Not included:

- SPTR (see I1 for provenance)
- Electronic jitter (TDC binning, cable, trigger)
- Optical crosstalk: 23% total at 12 V OV; spurious simultaneous pulse can advance the k th-order timestamp

Quoted $\sigma_{\text{TOP}} = 15.2 \text{ ps}$ is a lower bound.

Order-statistic caveat: $\sigma_{\text{SPTR}} / \sqrt{k \cdot N_{\text{active}}}$ applies to sample averages, not to the k th order statistic of the TOP estimator. Propagation of SPTR through the order- k estimator is pending formal derivation.

Current dataset (GEN-3):

- Branch: `feat/bar-end-vikuiti`
- Build: `build_end_vikuiti/`
- Macro: `scan_end_vik_sparse_top_v2`
- Files: `results/.../photon_hits_ej230_ntop{4,8,14,20}_x{-600,...,+600}.root`

Tree: `sipm_hits`
`event_id, face_type, local_id,
time_ns, pde, x_mm, gun_x_mm`
Analysis scripts:

- `optim/root_best_est/best_est_analysis.py`
- `presentation_v5/scripts/analysis_v5.py`
- `presentation_v6/scripts/ (v6)`

Key simulation parameters:

Parameter	Value
Events/file	5 000
Positions x	13 (-600 to $+600$ mm)
N_{TOP} configs	4, 8, 14, 20
Scintillator	EJ-230
Reflector	Vikuiti ESR ($R = 0.95$)
Geometry	GEN-3 (air gap + Mylar)

File	Description
v5_veff_fit.pdf	$\langle \Delta t \rangle$ vs x , linear fit (v_{eff})
v5_veff_residual.pdf	Nonlinearity residuals
v5_sigma_vs_x.pdf	$\sigma(x)$: global + oracle
v5_top_position_loo.pdf	TOP centroid LOO position
v5_pareto.pdf	N_{TOP} Pareto scan
fig1_kscan.pdf	k -scan at $x = 0$
fig_end_mscan.pdf	m -scan at $x = 0$
fig_ntop_scan_full.pdf	k -scan all positions
fig_npe_x.pdf	N_{pe} vs x , 3 materials
fig_sigma_t_x.pdf	σ_t vs x , 3 materials
fig_bulk_survival.pdf	Bulk attenuation survival
fig_survival_RN.pdf	Reflector bounce survival
figM1_mat_sigma_end.pdf	Material σ_{END} comparison
figM2_mat_npe.pdf	Material N_{pe} comparison